Model Diversity and Shared Blind Spots

An Efficient-Market Perspective on Combining Models

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Overview

- Combining models helps when they correct one another's mistakes. Can we expect commercially available models to supply this diversity?
- A provider can sell a useful combination as a new service. Competition may therefore absorb inexpensive improvements into the services already available.
- When accurate models cannot be improved by simple voting, their errors must concentrate on a common set of questions. More models need not supply complementary judgments.

When Does Combining Models Improve Accuracy?

A procedure combining several models' answers is an *ensemble*.
Different models may make different mistakes, so combining their answers can improve accuracy.
But if an inexpensive combination works better, a provider could sell it as a new service. Competition gives a reason that such improvements might already be available.

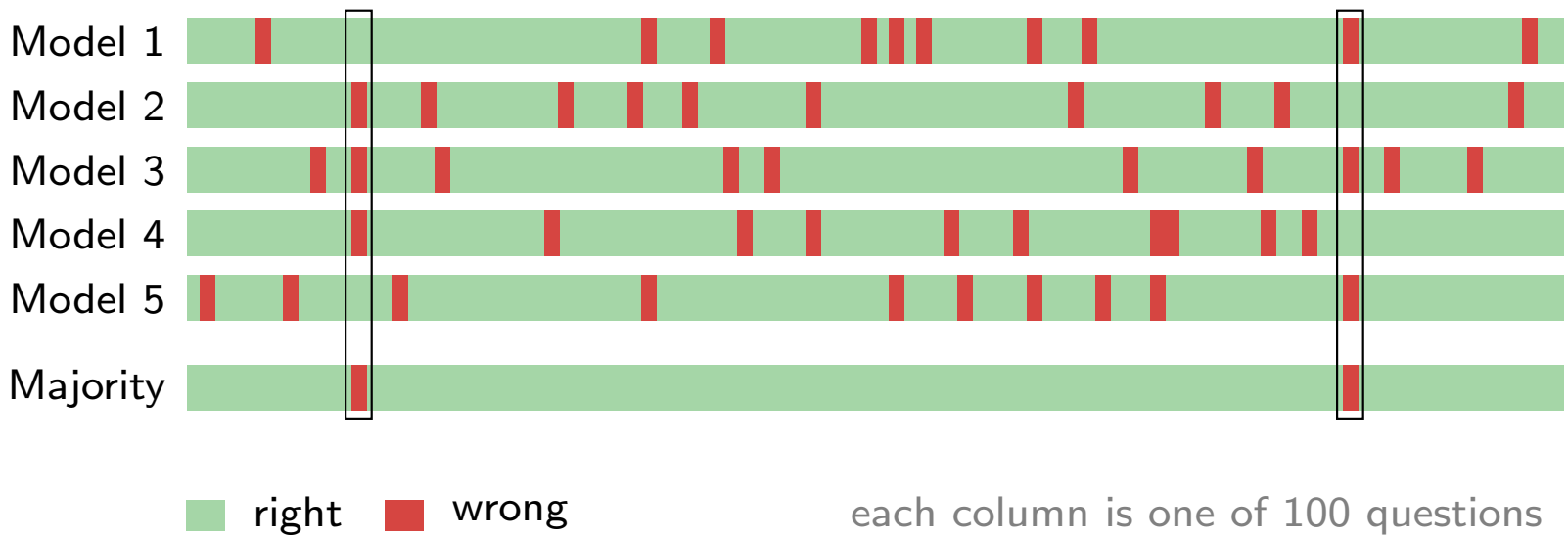
If simple combinations cannot outperform the best service at the same total cost, what does that imply about the models' mistakes?

This also matters when we assemble several models into an oversight panel.

Independent Errors Make Voting More Reliable

Suppose five binary predictors are each wrong on ten percent of questions from the same population. If their error indicators are mutually independent, majority vote errs only when at least three are wrong:

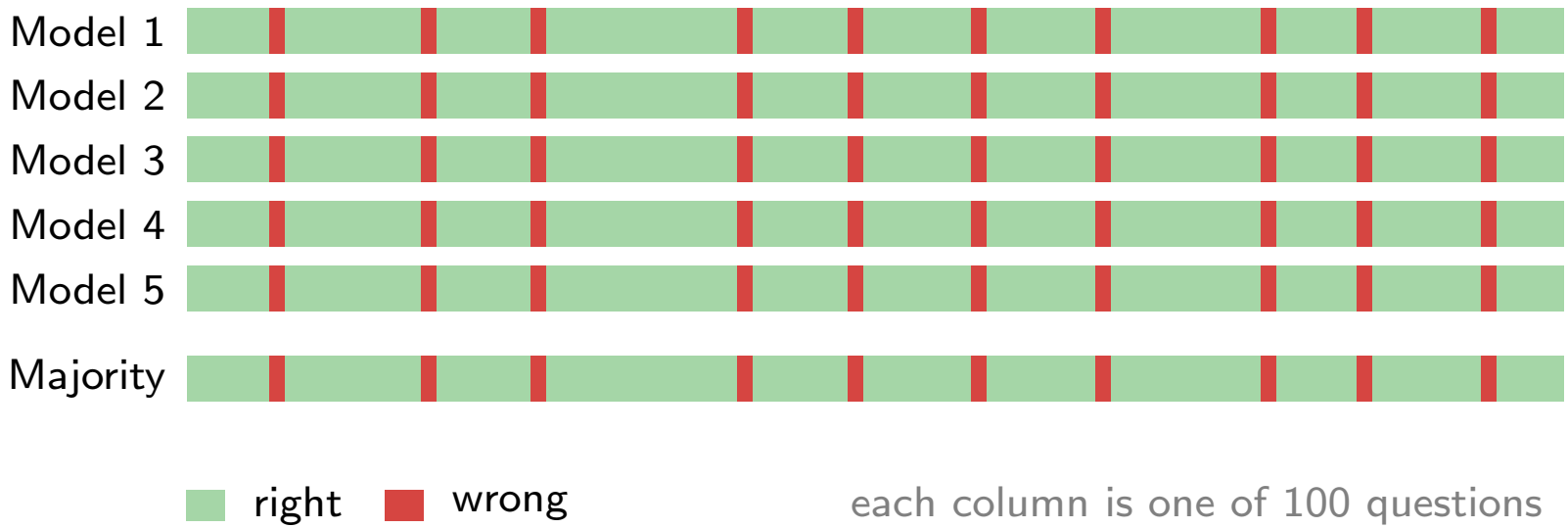
$$\sum_{j=3}^{5} \binom{5}{j} (0.1)^j (0.9)^{5-j} = 0.00856.$$



Errors scattered, ten per model. Boxed: the only two questions with three wrong answers.

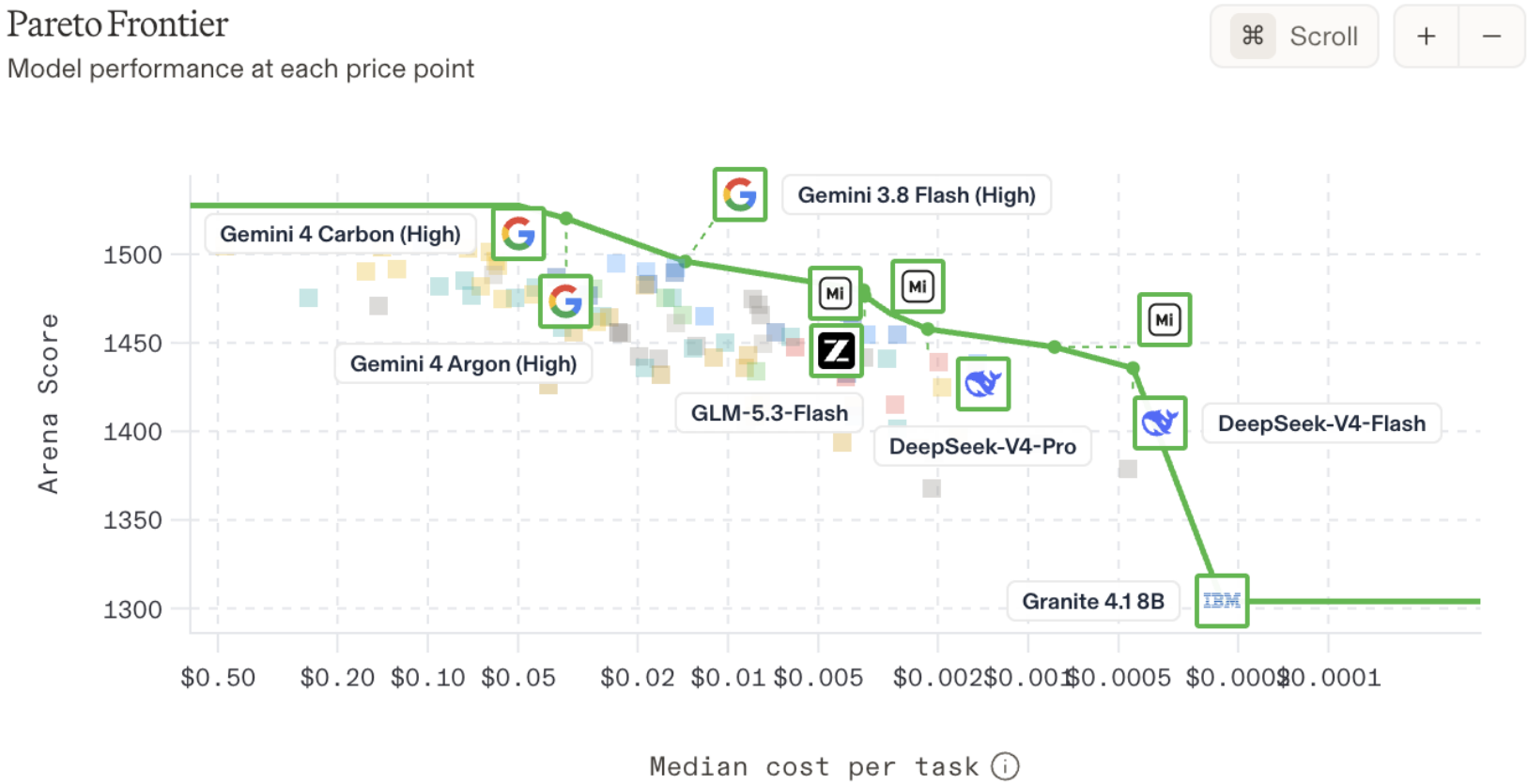
Identical Errors Leave Voting No Better

Now suppose all five predictors make exactly the same mistakes. Each is still wrong on ten percent of questions.



Majority vote errs on the same ten percent of questions.
The individual accuracies are identical in the two examples. What matters for voting is how the errors occur together.

What Performance Can We Buy at a Given Price?

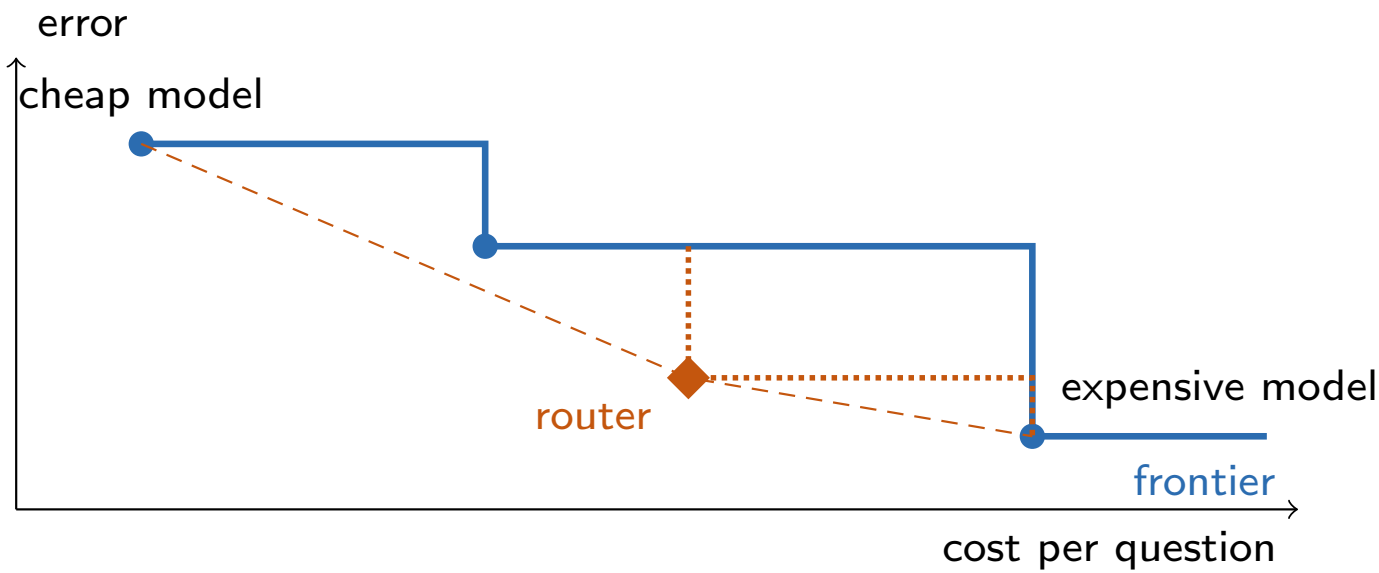


Higher is better; cheaper is to the right. Source: Arena.

The *Pareto frontier* consists of models for which no other listed model is at least as good and no more expensive, with one strict improvement.

A Better Ensemble Could Become a Market Service

An ensemble's benchmark is the best available service at its *total* cost, including every model call and the work of combining the answers. A *router* sends easy questions to a cheap model and hard ones to an expensive model.

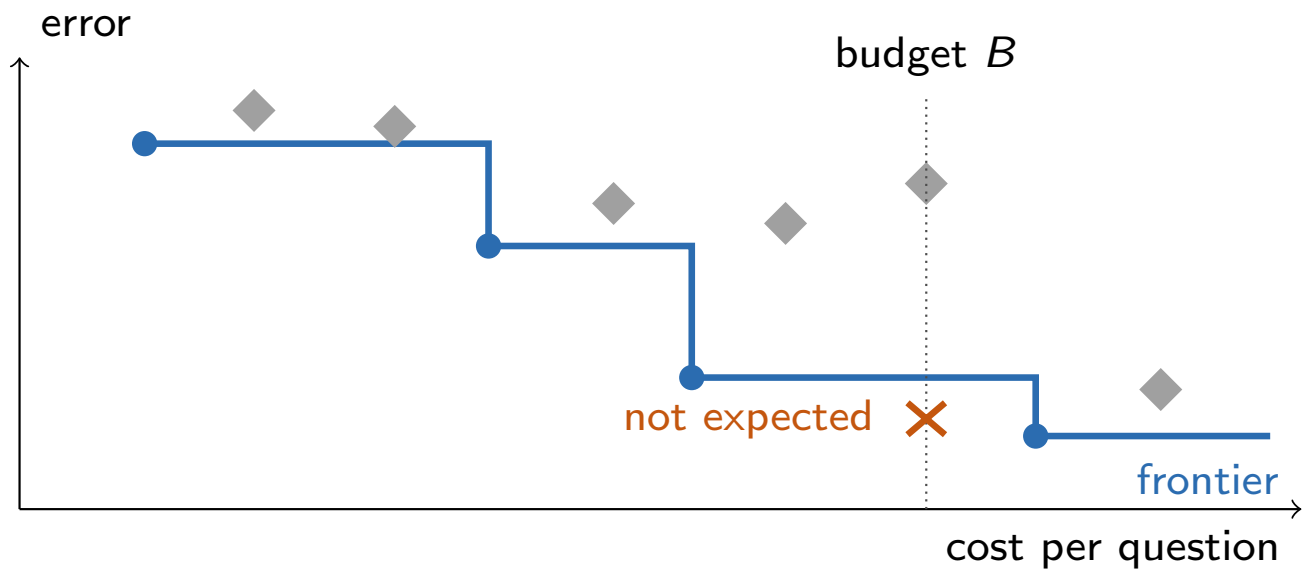


The router is more accurate than any service at its cost. A provider could sell it as a new service, and the frontier would move down (dotted), without training a new foundation model. Competitors could copy it.

The Efficient-Market Hypothesis for AI

Efficient-market hypothesis for AI

In a mature market, competition should tend to make simple, inexpensive ensemble improvements available as services buyers can already purchase.



Gray: simple ensembles such as votes and routers. None lands below the frontier. This is a motivation for our hypothesis, not a theorem about market dynamics.

Model: Binary Answers on a Fixed Population

For the analysis, replace Arena score by accuracy on a binary prediction task. Let $X \sim \mu$ be a question from a finite set $\mathcal{X}$, omitting questions with probability zero. Each input has a fixed, correct binary label

$$Y = y(X) \in \{-1, +1\}.$$

A model $h : \mathcal{X} \to \{-1, +1\}$ has population error

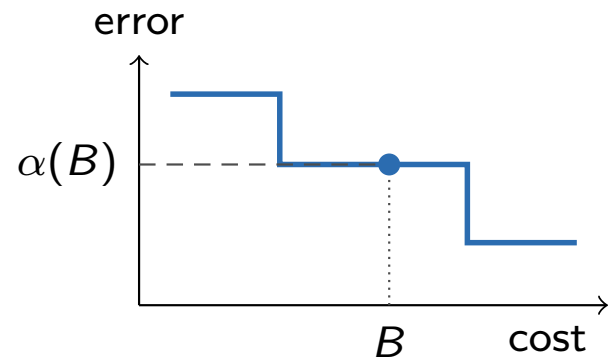
$$\mathrm{err}(h) := \mathbb{P}\big(h(X) \neq Y\big).$$

All probabilities and expectations are under $X \sim \mu$, unless another distribution is specified.
The models' answers are fixed; the random object is the question.
For oversight, an input is a proposal: its correct label is $+1$ if it does not harm the principal and -1 if it does.

What Accuracy Is Available at Budget B?

Fix a budget $B > 0$ per question. Before designing an ensemble, ask how accurately an existing service can answer for that price. A finite menu $\mathcal{M}$ contains services with fixed, nonnegative costs $c(h)$ per question. A service may itself combine models. Include the two free services that always predict $+1$ or always predict -1. Define the best available error at this budget:

$$\alpha(B) := \min_{\substack{h \in \mathcal{M} \\ c(h) \le B}} \operatorname{err}(h).$$



The budget is fixed throughout the analysis. Write $\alpha := \alpha(B)$ and consider the case $\alpha > 0$, where some errors remain.
The better constant makes error at most $1/2$, so

$$0 < \alpha \le \frac{1}{2}.$$

Threshold Votes Count Positive Answers

Choose $k \geq 2$ models $h_1, \ldots, h_k \in \mathcal{M}$ with total cost $\sum_j c(h_j) \leq B$. Call them *jointly affordable*.
Let $A(x)$ count how many models answer $+1$:

$$A(x) := \sum_{j=1}^{k} \mathbf{1}\{h_j(x) = +1\}.$$

For each threshold $q \in \{1, \ldots, k\}$, define the vote

$$v_q(x) := \begin{cases} +1, & A(x) \geq q, \\ -1, & A(x) < q. \end{cases}$$

The rule v_1 requires one positive answer; v_k requires unanimity.
Strict majority uses $q = \lfloor k/2 \rfloor + 1$.
The threshold stays fixed across questions. We idealize counting votes as free, so every v_q costs at most B.

Threshold Votes on Six Questions

Five models answer six questions. Shaded cells are $+1$.

	x_1	x_2	x_3	x_4	x_5	x_6
h_1	+	+	−	−	+	−
h_2	+	−	−	−	+	+
h_3	+	+	+	−	+	−
h_4	+	+	−	−	+	−
h_5	+	−	−	−	−	+
$A(x)$	**5**	**3**	**1**	**0**	**4**	**2**
v_1 (any)	+	+	+	−	+	+
v_2	+	+	−	−	+	+
v_3 (majority)	+	+	−	−	+	−
v_4	+	−	−	−	+	−
v_5 (unanimity)	+	−	−	−	−	−

The vote v_q says $+1$ exactly when at least q models do. Down each column, the votes are the models' answers sorted with $+1$ first.

A Panel Can Require a Fixed Number of Approvals

For a proposal x, encode reviewer j's report as $h_j(x) = +1$ for approval and -1 for rejection.
Requiring at least q approvals is exactly the rule v_q. Unanimous approval uses v_k.
A sincere report reflects the *reviewer's* preference; we evaluate it against the *human's* approve-or-reject decision. The report functions stay fixed while we vary the threshold.
Error counts both harmful approvals and non-harmful rejections. This is the accuracy question we study today.

Assume No Threshold Vote Beats the Budget Benchmark

Recall that $\alpha = \alpha(B)$ is the smallest error of an available service at budget B.

No improvement by threshold voting

For this jointly affordable collection,

$$\mathrm{err}(v_q) \geq \alpha \qquad \text{for every } q \in \{1, \ldots, k\}.$$

This assumes that *none* of the thresholds improves on the benchmark. Testing majority vote alone would not establish it. The conclusion will follow from the failure of just these simple rules. We need not assume that more sophisticated ensembles also fail.

How Often Can Any Model Be Wrong?

Write

$$E_j := \{x : h_j(x) \neq y(x)\}, \qquad U := \bigcup_{j=1}^{k} E_j.$$

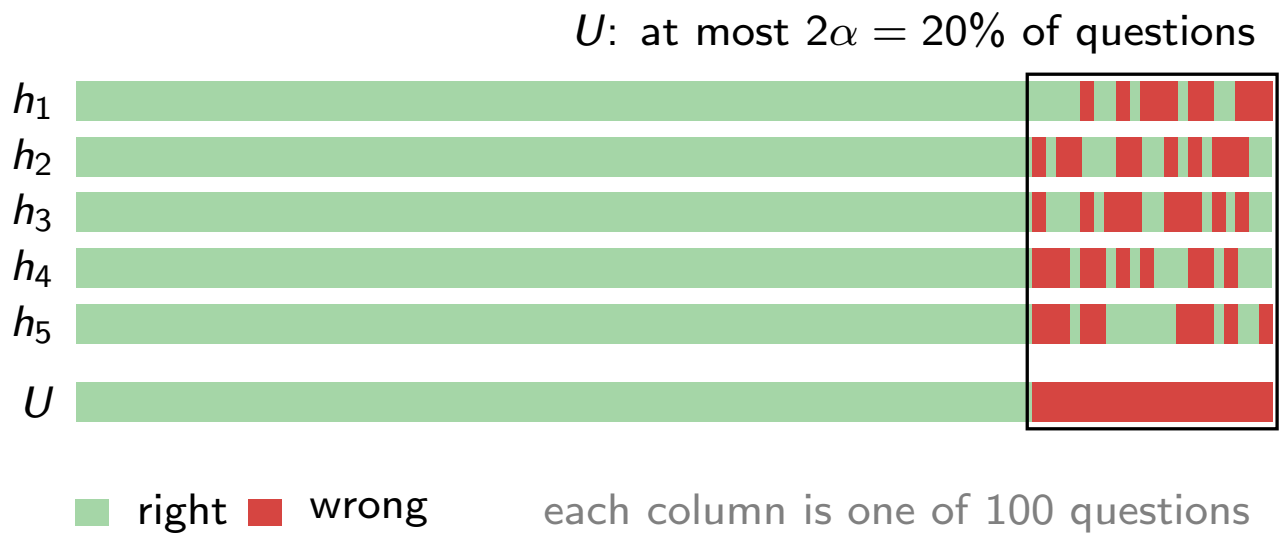
Thus U is the set of questions on which *at least one* model is wrong.
Let $\delta_j := \mathrm{err}(h_j) - \alpha$ be model j's *excess error* above the best service at budget B. Each model is itself affordable, so $\delta_j \geq 0$.

Theorem: Errors Concentrate

Theorem (Error concentration)

If every threshold vote of the jointly affordable models has error at least $\alpha > 0$, then

$$\mathbb{P}(U) \leq 2\alpha + \sum_{j=1}^{k} \delta_j.$$



Five models with error $\alpha = 10\%$ and no excess error. Questions are sorted so that U is on the right.

Threshold Votes Preserve the Error Count

Fix a question x and let $r = A(x)$ be its number of positive answers. The thresholds $q = 1, \ldots, r$ return $+1$; the other $k - r$ thresholds return -1.

	positive answers	negative answers
The k models	r	$k - r$
The k threshold votes	r	$k - r$

Whichever label is correct, both collections contain the same number of wrong answers on this question. Averaging this equality over questions gives

$$\boxed{\sum_{q=1}^{k} \mathrm{err}(v_q) = \sum_{j=1}^{k} \mathrm{err}(h_j).}$$

The Two Extreme Thresholds Detect Every Error

Recall that v_1 returns $+1$ if *any* model does, while v_k returns $+1$ only if *all* models do.
If $y(x) = +1$ and some model is wrong, at least one model answers -1. Unanimity fails, so $v_k(x) = -1$ is wrong.
If $y(x) = -1$ and some model is wrong, at least one model answers $+1$. Then $v_1(x) = +1$ is wrong.
Thus whenever $x \in U$, at least one of these two votes is wrong:

$$U \subseteq \{x : v_1(x) \neq y(x)\} \cup \{x : v_k(x) \neq y(x)\}.$$

The union bound gives

$$\boxed{\mathbb{P}(U) \leq \mathrm{err}(v_1) + \mathrm{err}(v_k).}$$

The Interior Thresholds Complete the Bound

The bound from the two extreme thresholds and the equality of total errors give

$$\begin{aligned}\mathbb{P}(U) &\leq \mathrm{err}(v_1) + \mathrm{err}(v_k) \\ &= \sum_{j=1}^{k} \mathrm{err}(h_j) - \sum_{q=2}^{k-1} \mathrm{err}(v_q).\end{aligned}$$

Each of the $k-2$ interior thresholds has error at least α. Using $\mathrm{err}(h_j) = \alpha + \delta_j$, we obtain

$$\begin{aligned}\mathbb{P}(U) &\leq \sum_{j} \mathrm{err}(h_j) - (k-2)\alpha \\ &= k\alpha + \sum_{j} \delta_j - (k-2)\alpha.\end{aligned}$$

$$\boxed{\mathbb{P}(U) \leq 2\alpha + \sum_{j} \delta_j.}$$

Small Excess Errors Give a Small Common Error Region

The theorem gives $\mathbb{P}(U) \leq 2\alpha + \sum_j \delta_j$, where $\delta_j = \text{err}(h_j) - \alpha(B)$. If every model matches the combined-budget benchmark, all $\delta_j = 0$:

$$\mathbb{P}(U) \leq 2\alpha.$$

For five models and $\alpha = 0.1$, all errors lie within at most 20% of questions. Every model is correct on the remaining 80% or more. The general bound adds the models' excess errors $\sum_j \delta_j$. It is informative when both that sum and α are small.
Equivalently, if $\mathbb{P}(U) > 2\alpha + \sum_j \delta_j$, at least one threshold vote must beat the best service at budget B. Sufficiently dispersed errors would allow an improvement.

Excess Error Includes the Gain from a Larger Budget

A model can be excellent for its own price while falling short of the best service at the ensemble's total budget.
For model h_j with price $c(h_j) \leq B$,

$$\begin{aligned}\delta_j &= \mathrm{err}(h_j) - \alpha(B) \\ &= \underbrace{\mathrm{err}(h_j) - \alpha(c(h_j))}_{\text{gap from the best at its own price}} + \underbrace{\alpha(c(h_j)) - \alpha(B)}_{\text{gain available with a larger budget}} .\end{aligned}$$

Both terms are nonnegative. Small excess error requires both good performance for the model's price and little improvement from spending the full budget.
When the frontier improves sharply with spending, the error-concentration bound can be weak even if every constituent is best at its own price.

Failures Occur Together on Average

Continue with $\text{err}(h_j) = \alpha$ for every model. The questions with any failure form $U = \bigcup_j E_j$, with $\mathbb{P}(U) \leq 2\alpha$. For each model,

$$\mathbb{P}(E_j \mid U) = \frac{\mathbb{P}(E_j)}{\mathbb{P}(U)} = \frac{\alpha}{\mathbb{P}(U)} \geq \frac{1}{2}.$$

Every model fails on at least half of the *same* subset of questions. Let $N(X) = \sum_j \mathbf{1}\{X \in E_j\}$ count the models that fail. Summing the conditional probabilities gives

$$\boxed{\mathbb{E}[N \mid U] \geq \frac{k}{2}.}$$

When any model fails, at least half fail *on average*. The models that fail can differ across questions.

Five Models Can Attain the Factor of Two

Consider this population of questions. Each row gives its probability, the correct label, and the models' answers.

Probability	Y	h_1	h_2	h_3	h_4	h_5
0.4	$+1$	$+$	$+$	$+$	$+$	$+$
0.4	-1	$-$	$-$	$-$	$-$	$-$
0.1	$+1$	$+$	$+$	$-$	$-$	$-$
0.1	-1	$+$	$+$	$-$	$-$	$-$

Models h_1, h_2 err only on the last row. Models h_3, h_4, h_5 err only on the third row. Thus every model has error 0.1, but

$$\mathbb{P}(U) = 0.2.$$

Could a voting rule improve on 0.1? The last two rows give it exactly the same answers.

The Same Answers Conceal Opposite Correct Labels

On the last 20% of questions, the available answers are

Probability	Y	h_1	h_2	h_3	h_4	h_5
0.1	$+1$	$+$	$+$	$-$	$-$	$-$
0.1	-1	$+$	$+$	$-$	$-$	$-$

Any fixed rule using only these answers makes the same prediction on both rows. It is wrong on at least one row, giving error at least 0.1.

Take a market containing these five jointly affordable models and the constants. Then $\alpha = 0.1$, no threshold vote improves on it, and $\mathbb{P}(U) = 2\alpha$. The bound is tight.

The models' mistakes concentrate in a small common region, even though h_1 and h_3 never fail on the same question.

Weighted Votes Generalize Threshold Votes

The bound $2\alpha + \sum_j \delta_j$ grows with the models' excess errors. Can a stronger assumption remove that term?

A *weighted vote* fixes real weights $a_1, \ldots, a_k$ and an offset b, and predicts

$$F(x) := \operatorname{sign}\big(g(x)\big), \qquad g(x) := b + \sum_{j=1}^{k} a_j \, h_j(x).$$

A score $g(x) = 0$ counts as an error.

- Equal weights and $b = 0$ give majority vote.
- Weights $a_j = 1$ and offset $b = k - 2q + 1$ give the threshold vote v_q.
- A negative weight reverses a model's answers.

No improvement by weighted voting

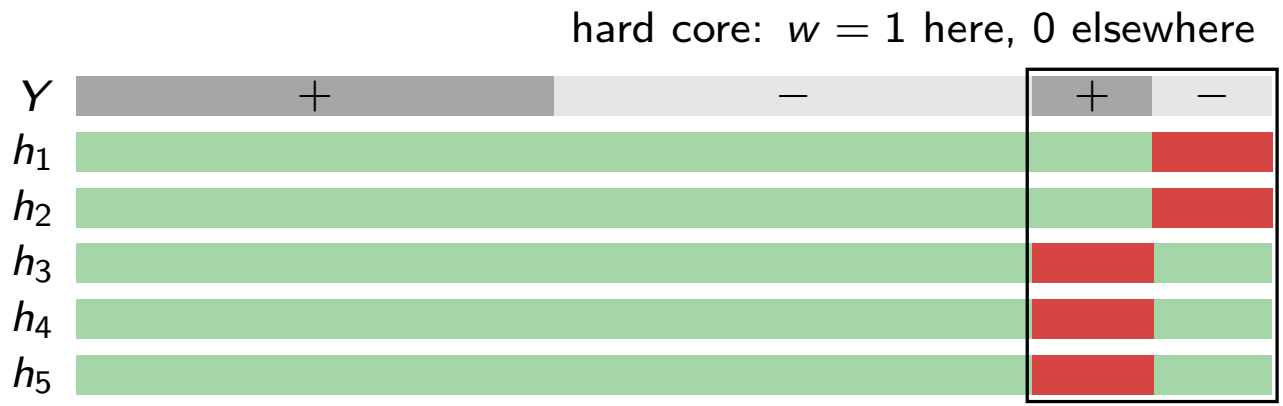
Every weighted vote of the jointly affordable models has error at least α.

A Hard Core: Where Every Model Is at Chance

Give each question x a weight $w(x) \in [0, 1]$, and call $\mathbb{E}[w(X)]$ the size of the weighting. Then draw questions in proportion to w.

Definition (Hard core)

A weighting w of positive size is a *hard core* if, under it, the two labels are equally likely and every model is right exactly half the time.



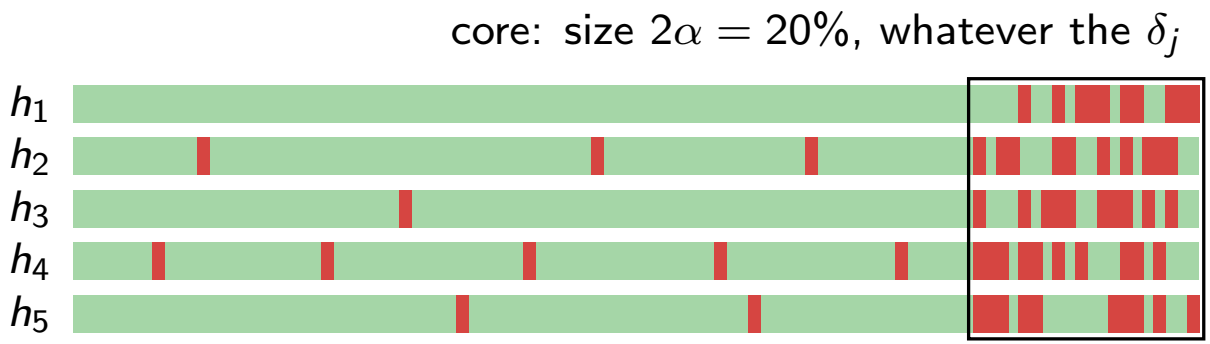
The five-model example from before. In the last 20%, half the labels are $+1$ and each model is right on half.

Theorem: A Common Hard Core of Size 2α

Theorem (Common hard core)

If every weighted vote of the jointly affordable models has error at least $\alpha > 0$, then the models have a hard core of size 2α.

Each model is right on half the core, so the core holds α of each model's error. The rest, $\text{err}(h_j) - \alpha = \delta_j$, lies outside.



Schematic with $\alpha = 10\%$. Red cells outside the box are excess errors. In general a core may weight questions partly.

Proof Sketch, Step 1: Cores Are Bounded by Hinge Loss

Write the *signed score* of a weighted vote as $m(x) := y(x)g(x)$.
The vote is right exactly when $m(x) > 0$.
On a hard core, $\mathbb{E}[wY] = 0$ (balanced labels) and $\mathbb{E}[wYh_j] = 0$ (each model at chance). Since $Yg = bY + \sum_j a_j Yh_j$, every weighted vote has

$$\mathbb{E}\big[w\, Yg\big] = 0.$$

Using $0 \le w \le 1$, for any core w and any score g,

$$\mathbb{E}[w] = \mathbb{E}\big[w(1 - Yg)\big] \le \mathbb{E}\big[(1 - Yg)_+\big] =: \mathrm{hinge}(g).$$

So no core is larger than the hinge loss of any vote. The next slide shows the two are equal at their optima.

Proof Sketch: The Core Is a Linear Program

Each expectation is a finite sum, so both are linear programs.

Largest core

$$\begin{aligned} \max_{w} \quad & \mathbb{E}[w(X)] \\ \text{s.t.} \quad & 0 \le w(x) \le 1 \quad \forall x \\ & \mathbb{E}[wY] = 0 \\ & \mathbb{E}[wYh_j] = 0 \quad \forall j \end{aligned}$$

Smallest hinge loss

$$\begin{aligned} \min_{b,a,\xi} \quad & \mathbb{E}[\xi(X)] \\ \text{s.t.} \quad & \xi(x) \ge 0 \quad \forall x \\ & \xi(x) \ge 1 - y(x)g(x) \ \forall x \end{aligned}$$

where $g = b + \sum_j a_j h_j$.

Constraint on w	Its multiplier in the dual
$\mathbb{E}[wY] = 0$ (balanced labels)	offset b
$\mathbb{E}[wYh_j] = 0$ (model j at chance)	weight a_j
$w(x) \le 1$	slack $\xi(x)$

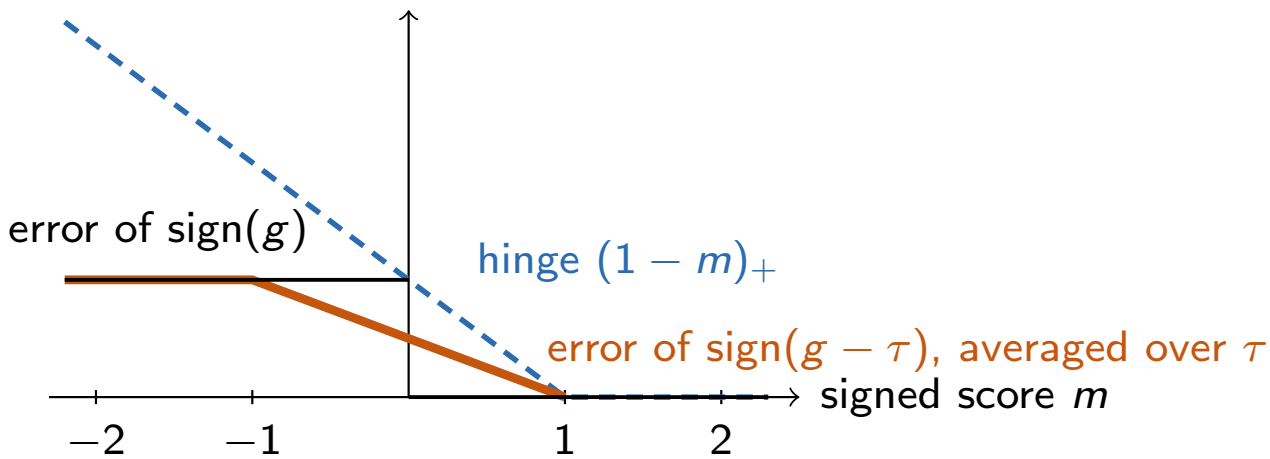
At the optimum $\xi = (1 - yg)_+$, so the dual value is the smallest hinge loss. The core LP is feasible ($w = 0$) and bounded, so strong duality makes the two optima equal.

Proof Sketch, Step 2: We Need Error, Not Hinge Loss

So far: the largest core equals the smallest hinge loss, $\mathrm{hinge}(g)$, over weighted votes. No-arbitrage is a statement about *error*. So it is enough to find a weighted vote with error at most $\mathrm{hinge}(g)/2$. Then $\alpha \le \mathrm{hinge}(g)/2$, and the largest core is at least 2α. A first try: wherever $\mathrm{sign}(g)$ is wrong, the signed score is $m \le 0$, so $(1-m)_+ \ge 1$. Hence $\mathrm{err}(\mathrm{sign}\, g) \le \mathrm{hinge}(g)$. This only gives a core of size α. To gain the factor of two, keep the weights and move the threshold: predict $\mathrm{sign}(g - \tau)$ for a number τ. This is still a weighted vote, with offset $b - \tau$, so no-arbitrage applies to it.

Proof Sketch, Step 3: A Random Threshold

Draw τ uniformly from $[-1, 1]$. At a question with signed score m, the vote $\text{sign}(g - \tau)$ is wrong when $y\tau \geq m$. Since $y\tau$ is uniform on $[-1, 1]$, that has probability at most $(1 - m)_+/2$.

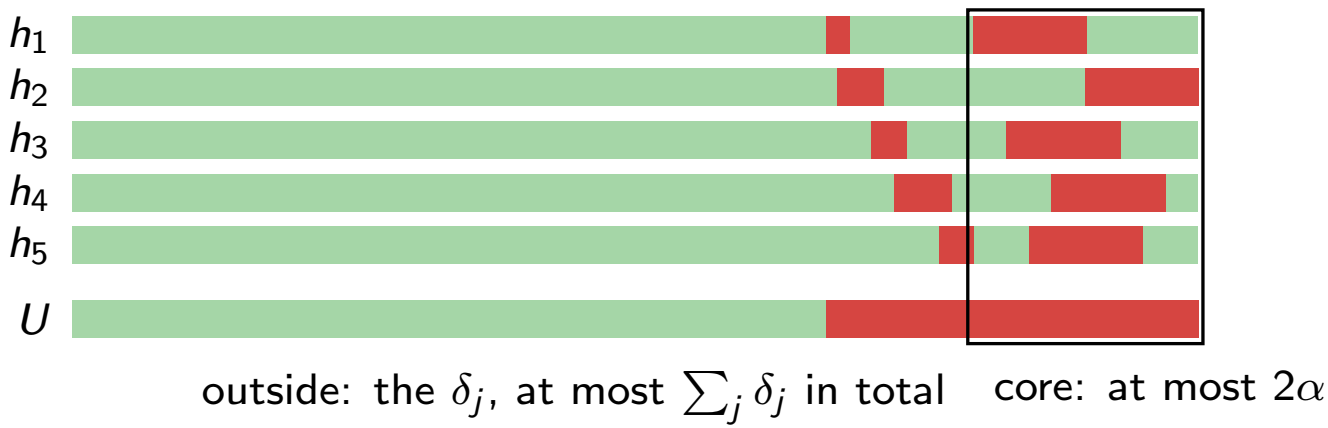


Averaged over questions and τ, the error is at most $\text{hinge}(g)/2$, so some fixed τ does as well. Hence $\alpha \leq \text{hinge}(g)/2$: the largest core is at least 2α. Scale it down to exactly 2α.

The Hard Core Recovers Error Concentration

The hard core is not a weaker notion: it implies the bound we proved earlier. Take a core w of size 2α and split $\mathbb{P}(U)$:

$$\mathbb{P}(U) = \underbrace{\mathbb{E}\big[w\,\mathbf{1}_U\big]}_{\leq \mathbb{E}[w] = 2\alpha} + \underbrace{\mathbb{E}\big[(1-w)\,\mathbf{1}_U\big]}_{\leq \sum_j \delta_j} \leq 2\alpha + \sum_{j=1}^{k} \delta_j.$$



Each model has exactly α of its error in the core, so δ_j outside it. The union bound adds these up. The same bound as before, from a stronger assumption.

Summary

- Competition gives a reason that simple ensemble improvements might already be available as services. Compare a new combination with what buyers can get at its *total* cost.
- If accurate models cannot improve on that benchmark by threshold voting, their errors concentrate on a small common set of questions. The bound depends on their excess errors at the full budget.
- If weighted votes cannot improve either, the models share a hard core of size 2α on which every one of them is at chance.
- Buying more models does not automatically buy complementary judgments. An oversight mechanism can use such complementarity, but establishing it is a separate design problem.

Thanks!

See you next class!

Optional: Averaging Probability Forecasts

The same market question applies when models report probabilities. Now a forecast is a vector

$$p(x) = (p_-(x), p_+(x)).$$

Its coordinates are nonnegative and sum to one. They give the reported probabilities of labels $-1, +1$.
Let e_Y be the point-mass vector at the correct label: $e_{-1} = (1, 0)$ and $e_{+1} = (0, 1)$. The *Brier loss* is

$$L(p) := \mathbb{E}\big[\|p(X) - e_Y\|_2^2\big].$$

Two forecasts can be combined by reporting

$$\bar{p}(x) := \frac{p_1(x) + p_2(x)}{2}.$$

How different can p_1, p_2 be if their average cannot improve on the best available service?

No Improvement by Averaging Bounds Disagreement

We now compare services by Brier loss, rather than classification error. Let β be the smallest Brier loss of an available forecasting service at budget B.
Choose two services p_1, p_2 whose combined cost is at most B.
Averaging is free. Define their excess losses

$$\varepsilon_j := L(p_j) - \beta \geq 0, \qquad j = 1, 2.$$

Assume their average cannot beat the benchmark: $L(\bar{p}) \geq \beta$.

Theorem (Forecast agreement)

Under these assumptions,

$$\boxed{\mathbb{E}\|p_1(X) - p_2(X)\|_2^2 \leq 2(\varepsilon_1 + \varepsilon_2).}$$

The proof will identify exactly how much averaging reduces Brier loss.

The Exact Gain from Averaging Forecasts

Fix a question and write $a = p_1(x) - e_{y(x)}$, $b = p_2(x) - e_{y(x)}$ for the two forecast errors. Their average is the error of $\bar{p}$. Expanding the squared norms,

$$\frac{\|a\|_2^2 + \|b\|_2^2}{2} - \left\|\frac{a+b}{2}\right\|_2^2 = \frac{\|a\|_2^2 + \|b\|_2^2 - 2\langle a, b\rangle}{4}$$
$$= \frac{1}{4}\|a - b\|_2^2.$$

Since $a - b = p_1(x) - p_2(x)$, taking expectations gives

$$\boxed{L(\bar{p}) = \frac{L(p_1) + L(p_2)}{2} - \frac{1}{4}\mathbb{E}\|p_1(X) - p_2(X)\|_2^2.}$$

The squared disagreement measures the improvement over the *average loss of the two models*.

Forecasts Near the Benchmark Must Be Close

Using $L(p_j) = \beta + \varepsilon_j$ in the averaging identity gives

$$L(\bar{p}) = \beta + \frac{\varepsilon_1 + \varepsilon_2}{2} - \frac{1}{4}\mathbb{E}\|p_1(X) - p_2(X)\|_2^2.$$

The no-improvement hypothesis $L(\bar{p}) \geq \beta$ therefore implies

$$\frac{1}{4}\mathbb{E}\|p_1(X) - p_2(X)\|_2^2 \leq \frac{\varepsilon_1 + \varepsilon_2}{2},$$

which proves the bound.
If both excess losses are at most ε, then

$$\mathbb{E}\|p_1(X) - p_2(X)\|_2^2 \leq 4\varepsilon.$$

At zero excess loss, the forecasts agree on every positive-probability question, even if the benchmark loss β is large. Substantial disagreement would let us beat the benchmark by averaging these near-benchmark forecasts.

References

- Surbhi Goel, Meena Jagadeesan, Michael Kearns, Sophia Pi, and Aaron Roth. *Ensemble Arbitrage and the Efficient Market Hypothesis for AI*. Unpublished manuscript, 2026. Section 3.2 gives the threshold-voting proof and tight example; Section 5.2 and Appendix C give the hard core for weighted votes; Appendix A gives the market application for Brier loss.
- Arena. Text leaderboard, Pareto view. Cost–performance screenshot supplied for this lecture.

References: Forecast Averaging

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